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| **PI Name:** | Bayuzick, Robert J. Ph.D. | | |
| **Project Title:** | Investigation of the Relationship between Undercooling and Solidification Velocity | | |
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| **City:** | Nashville | **State:** | TN |
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| **COI Name (Institution):** | Hofmeister, William ( Vanderbilt University ) | | |
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Containerless processing is inherently necessary to obtain accurate measures of the solidification velocity as a function of undercooling. The absence of a container promotes high degrees of undercooling in a melt and allows the application of direct imaging techniques to obtain accurate measures of solidification velocity as a function of undercooling. For this work, electromagnetic levitation and electrostatic levitation are the appropriate techniques. In a previous project, electromagnetic levitation techniques were already extensively used on a variety of alloys. However, electromagnetic levitation is highly perturbing, producing liquid flows of the order of tens of centimeters per second. As a result, it is not entirely clear that experimental results are unaffected. Since electrostatic levitation is a more benign technique, experiments are to be duplicated and compared to electromagnetic levitation. A basis for low earth orbit experimentation, the most benign experiments, will be evaluated upon a comparison of results. At every step, experimental results are to be examined within the theoretical framework. This work is aimed at reconciling the

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| **Task Description:** | differences between experimental measurements of the theoretical predictions of the solidification velocity as a function of undercooling. To date, the theory proposed by Boettinger, Coriell and Trivedi (the BCT theory) has been one of the most widely used models for describing the nature of the solidification of undercooled metals and alloys. However, for undercoolings greater than about 5% of the absolute melting temperature, there is considerable discrepancy between theory and experiment. At these large undercoolings, experimental results exhibit a much lessened dependency of solidification velocity on undercooling than is predicted by theory. Furthermore, unpredicted plateaus in the solidification velocity as a function of undercooling are observed. The understanding of solidification kinetics is fundamental to the synthesis of materials. The understanding of the speed of solidification, along with nucleation frequencies, are missing links for modelers who are focused on the development of innovations in solidification processing. Basically, knowing the manner in which to control and predict the speed of solidification is the key to knowing how to influence the bulk microstructure and, as a result, the properties and performance of the material through solidification processing. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | Electrostatic Levitation Experiments<br>Solidification velocity measurements in electrostatic levitation (ESL) were completed for a wide range of undercoolings for Ni-4at.%Sn in FY04. A comparison between electromagnetic levitation (EML) and ESL was made since these two experimental techniques cause different streamline flows in the sample [1], which has an unknown effect on solidification velocity. The results of ESL-processing show solidification velocity results to be similar to those obtained by Altgilbers et al. [2] in EML processing. This demonstrates that solidification velocity is unaffected by turbulent streamline flow in EML when compared to the laminar streamline flow in ESL for the range of velocities measured in the experiment.<br><br>Phase Separation in Ni-4at.%Sn<br><br>In the process of the investigation of solidification velocity in ESL, a solid-state phase transformation was discovered in Ni-4at.%Sn. After rapid solidification of this alloy, phase separation occurs at 1630 K yielding two phases: a tin-rich phase (approximately 7.5at.%Sn) and a tin-deficient phase (approximately 3at.%Sn). It is believed that the phase separation occurs via a nucleation and growth process with an enthalpy of separation of approximately 3 kJ/mol. The microstructure is similar to that found in spinodal decomposition, but further experimentation would be necessary to determine the location of the chemical spinodal in the Ni-Sn phase diagram. X-ray diffraction (XRD) shows both phases to be of the A1 crystal structure, but electron backscatter patterns (EBSP; also backscatter Kikuchi patterns or BKP) will be obtained at Oak Ridge National Laboratories in order to establish orientation relationships between the two phases.<br><br>Phase separation can occur during EML processing of Ni-4at.%Sn, but whether phase separation occurs depends on the sample's cooling rate. High cooling rates after rapid solidification (greater than 10 K/s) suppress the phase separation, preserving the dendritic microstructure obtained by rapid solidification. Therefore, the final microstructure of EML-processed Ni-4at.%Sn can be controlled by simply controlling the flow rate of the cooling gas inside the levitator. Ni-4at.%Sn samples cast in an arc-melter or induction furnace, which are subject to high cooling rates, do not exhibit the phase separation.<br><br>Effect of Residual Gases on Solidification<br><br>Residual gas analyses on several nickel and Ni-4at.%Sn samples were completed. One nickel sample processed in EML using a He-15vol%H2 cooling gas was analyzed for helium content by a vaporization/mass spectrometry technique [3,4] at Battelle. The results showed that EML-processing does not significantly increase the helium content in nickel. Ni-4at.%Sn samples processed in ESL were analyzed by HORIBA (a combustion analysis similar to LECO but better suited to the small sample sizes used in ESL). The hydrogen, oxygen and nitrogen concentrations measured in these samples were similar to the results obtained by Algoso et al. [5] for Ni-5at.%Cu and Ni-10at.%Cu in EML processing. In general, nitrogen concentrations were below detectable limits, oxygen concentrations were on the order of 150 ppm O and the average hydrogen concentration was 350 ppm H.<br><br>It was previously hypothesized that oxygen played a role in reducing solidification velocity at high undercoolings [6]. However, based on the results reported in the FY03 task progress and the new results described here, oxygen plays a small role in the solidification velocity of rapidly solidified nickel and nickel-based alloys. Hydrogen is most likely the cause of historically observed deviations from theoretical predictions of solidification velocity at high undercoolings. Modeling of solidification velocity using the BCT theory [7] and thermophysical parameters obtained by computational thermodynamics [8] strongly supports this hypothesis. These results were submitted for publication in Scripta Materialia.<br><br>Modeling of experimental results<br><br>As previously reported, an alternate theory or "selection criterion" (the optimum stability conjecture or OSC [9]), was being considered for modeling experimental solidification velocity results. In FY04, low undercooling experiments were conducted on Ni-1.7at.%C, Ni-1at.%Si, Ni-5at.%Ti and Ni-5at%Cu in EML. The results support OSC and were presented at the TMS Annual Meeting 2004.<br><br>==<br><br>1. Flemings, M.C., Matson, D.M., Hyers, R.W. and Rogers, J.R., "Flight Planning for the International Space Station – Levitation Observation of Dendrite Evolution in Steel Ternary Alloy Rapid Solidification (LODESTARS)," Proceedings of the 5th NASA Microgravity Materials Science Conference, Huntsville, AL, NASA/CP-2003-212339, Gilles D., Ramachandran, N., Murphy K., McCauley, D. and Bennett N. (eds.) p. 221.<br><br>2. Altgilbers, A.S., Hofmeister, W.H. and Bayuzick, R.J., "The Dendrite Growth Kinetics of Nickel-based Alloys," Mater. Sci. Eng. A 360 (2003) 26.<br><br>3. Farrar, H. and Oliver, B.M., "A Mass Spectrometer System to Determine Very Low Levels of Helium in Small Solid and Liquid Samples," J. Vac. Sci. Technol. A 3 (1986) 1740.<br><br>4. Oliver, B.M., Bradley, J.G. and Farrar H., "Helium Concentration in the Earth's Lower Atmosphere," Geochim. Cosmochim. Acta 48 (1984) 1759. |

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| | 5. Algoso, P.R., Hofmeister, W.H. and Bayuzick, R.J., “Solidification Velocity of Undercooled Ni-Cu Alloys,” Acta Mater. 51 (2003) 4307.<br><br>6. Bassler, B.T., Hofmeister, W.H. and Bayuzick, R.J., “The Solidification Velocity of Pure Nickel,” Mater. Sci. Eng. A 342 (2003) 80.<br><br>7. Boettinger, W.J., Coriell, S.R. and Trivedi, R., “Application of Dendritic Growth Theory to the Interpretation of Rapid Solidification Microstructures,” Rapid Solidification Processing IV: Principles and Technologies, ed. R. Mehrabian and P.A. Parrish (Baton Rouge, LA: Claitor’s Publishing Division, 1988), 13.<br><br>8. Zeng, K., Klassen, T., Oelerich, W. and Bormann, R., “Thermodynamics of the Ni-H System,” J. Alloys Comp. 283 (1999) 151.<br><br>9. Sekerka, R.F., “Optimum Stability Conjecture for the Role of Interface Kinetics in Selection of the Dendrite Operating State,” J. Cryst. Growth 154 (1995) 337. |
| **Bibliography Type:** | Description: (Last Updated: 03/14/2008) |
| **Dissertations and Theses** | Algoso, P.R. "Rapid solidification of nickel and nickel-based alloys" Ph.D. Dissertation, Vanderbilt University, Nashville, TN<br>May-2004 |
| **Papers from Meeting Proceedings** | Algoso, P.R.; Hofmeister, W.H.; Bayuzick, R.J. "Optimum stability in rapidly solidified nickel-based alloys" Solidification Processes and Microstructures: A Symposium in Honor of Wilfried Kurz<br>Mar-2004 |
| **Presentation** | Algoso, P.R.; Hofmeister, W.H.; Bayuzick, R.J. "Residual gas effects on solidification velocity in electromagnetic levitation" ASM Materials Solutions 2003<br>Oct-2003 |
| **Presentation** | Algoso, P.R.; Hofmeister, W.H.; Bayuzick, R.J. "Optimum stability in rapidly solidified nickel-based alloys" TMS Annual Meeting 2004<br>Mar-2004 |